MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: RECEPTACLES, 90 THROUGH 240 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

This specification sheet is approved for use by Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

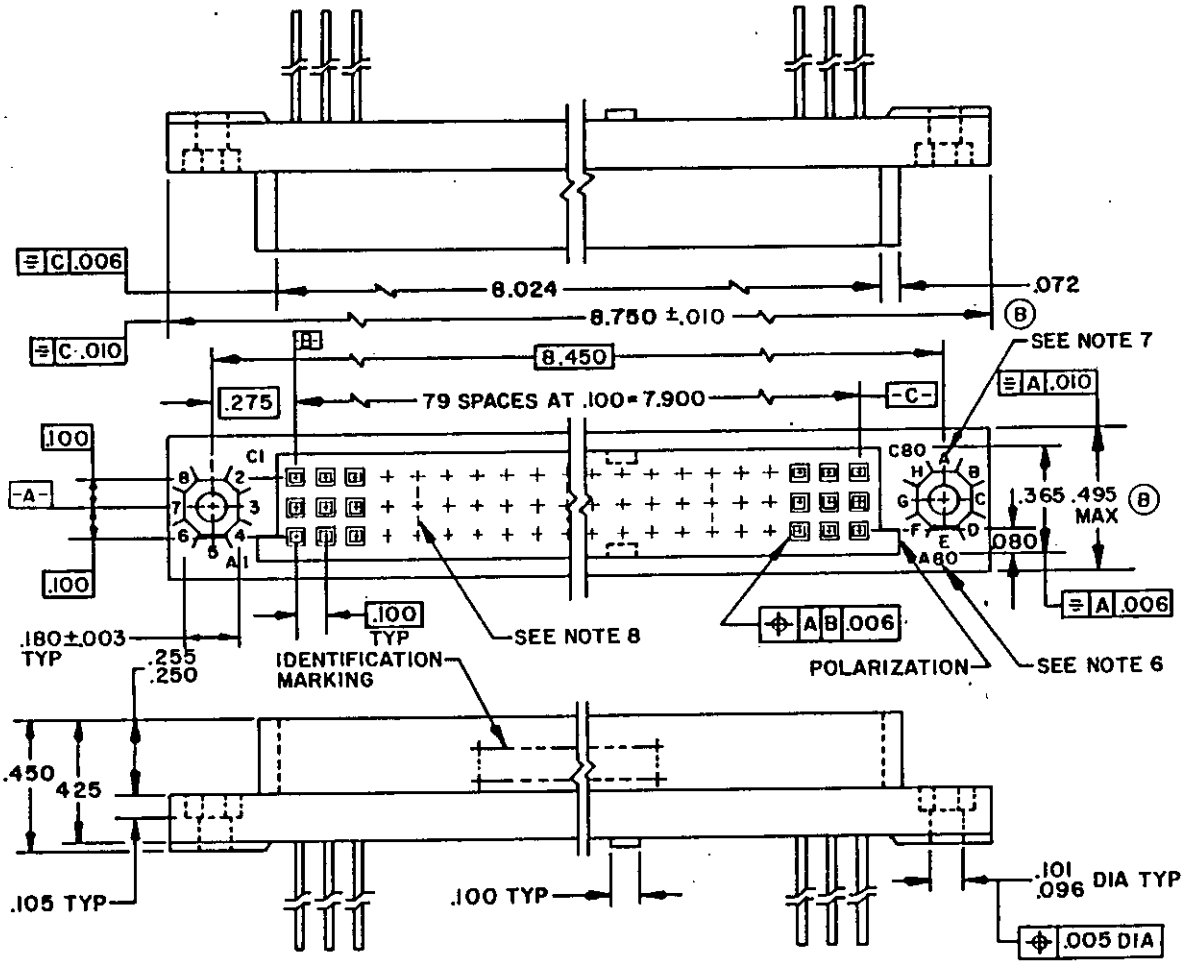
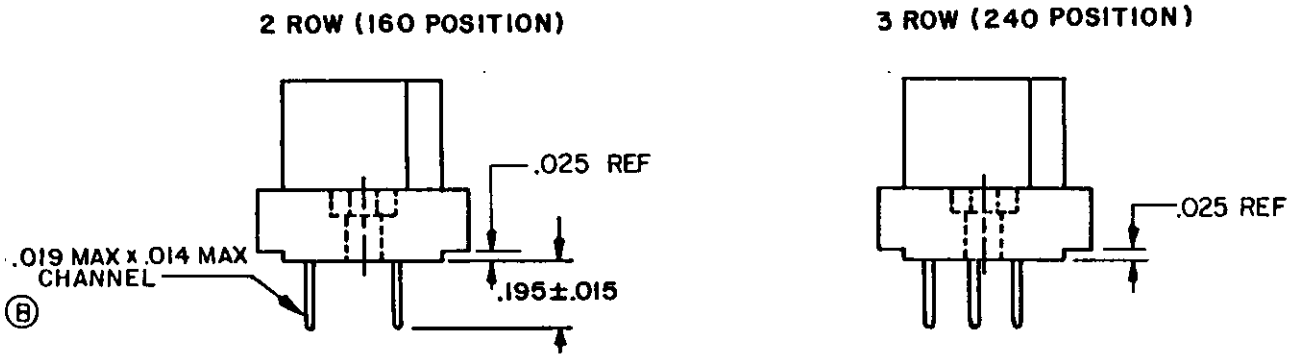


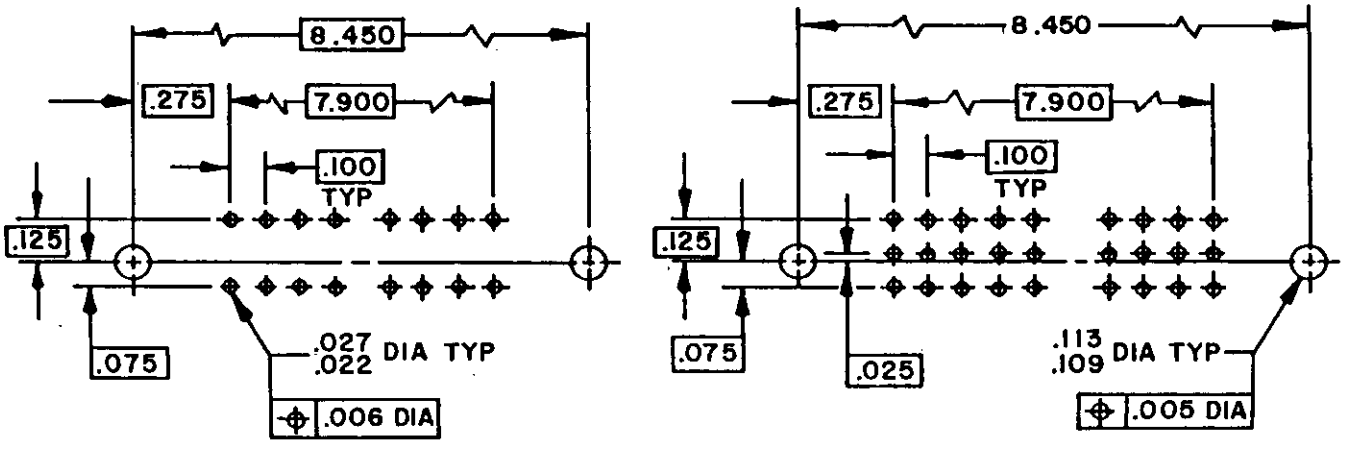
FIGURE 1. Connectors, kit receptacles (.100 spacing) 160 and 240 contacts.

Ⓑ denotes change

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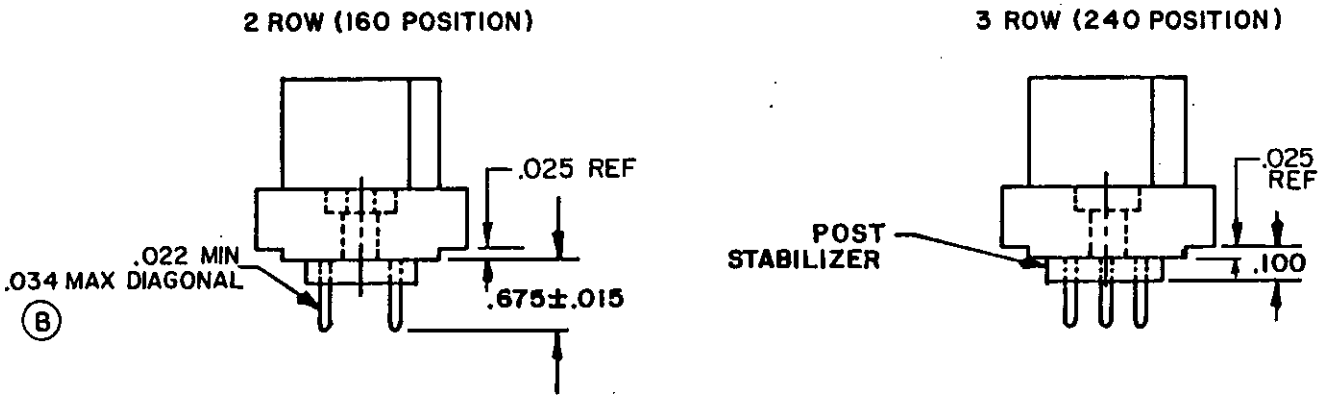
Contact type 1



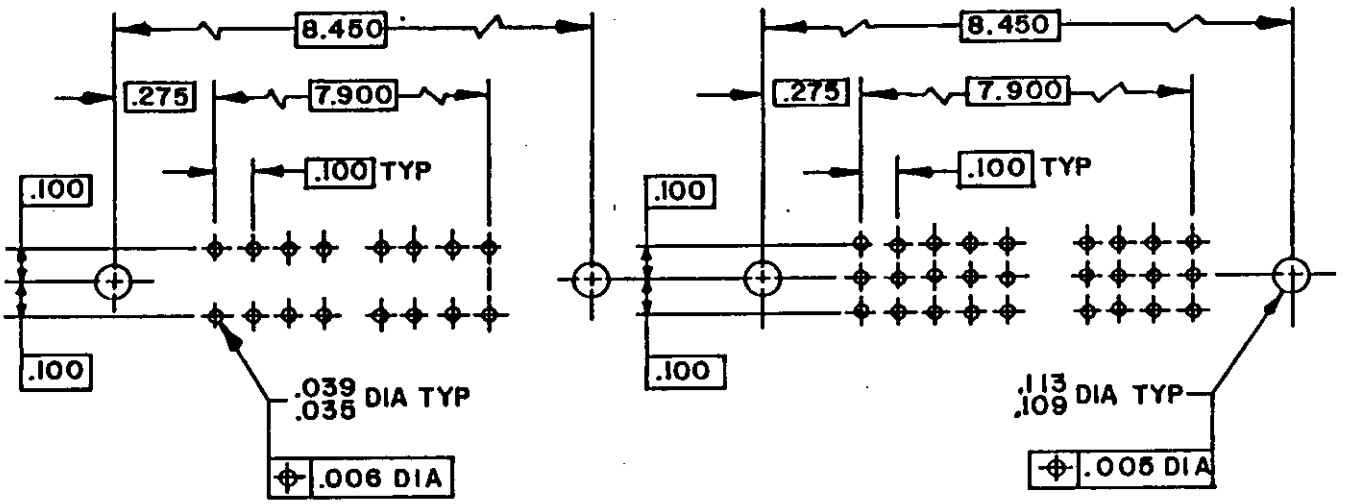
Contact types and recommended circuit layouts

FIGURE 1. Connectors, kit receptacles (.100 spacing) - 160 and 240 contacts - Continued.

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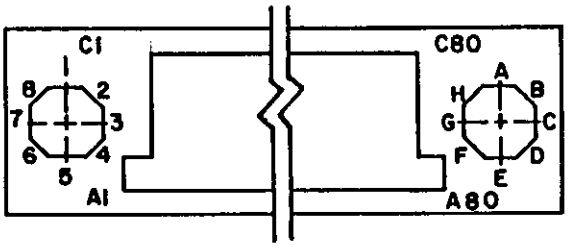
Contact type 2



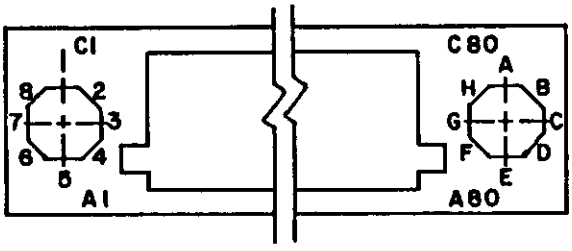
Contact types and recommended circuit layouts

FIGURE 1. Connectors, kit receptacles (.100 spacing) - 160 and 240 contacts - Continued.

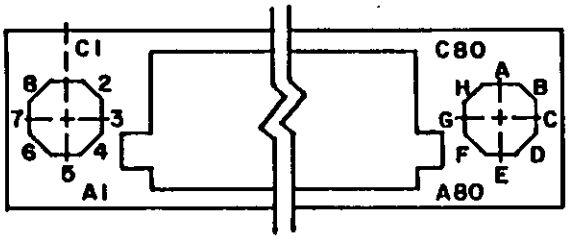
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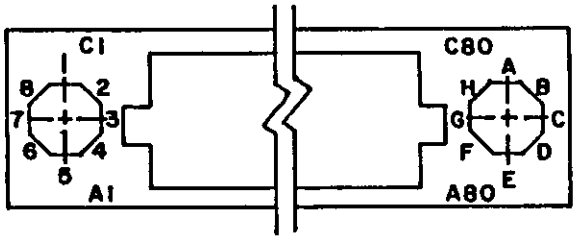
TYPE I



TYPE II



TYPE III



TYPE IV

POLARIZATION

Inches	mm	Inches	mm	Inches	mm
.003	0.08	.039	0.99	.255	6.48
.005	0.13	.072	1.83	.275	6.99
.006	0.15	.073	1.85	.365	9.27
.010	0.25	.080	2.03	.425	10.80
.014	0.36	.096	2.44	.450	11.43
.015	0.38	.100	2.54	.495	12.57
.019	0.48	.109	2.77	.675	17.15
.022	0.56	.113	2.87	7.700	200.66
.027	0.69	.180	4.57	8.024	203.81
.034	0.86	.195	4.95	8.450	214.63
.035	0.89	.250	6.35	8.750	222.25

(B)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified; tolerance is ±.005 (0.13 mm) on decimals and +2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/129.
5. Use printed circuit boards with both the type 1 and type 2 receptacles.
6. Numbers indicating end cavities and letters indicating rows stamped this surface.
7. Location indicators embossed on surface.
8. Lines indicate every five cavities stamped on mating face except end cavities.

FIGURE 1. Connectors, kit receptacles (.100 spacing) - 160 and 240 contacts - Continued.

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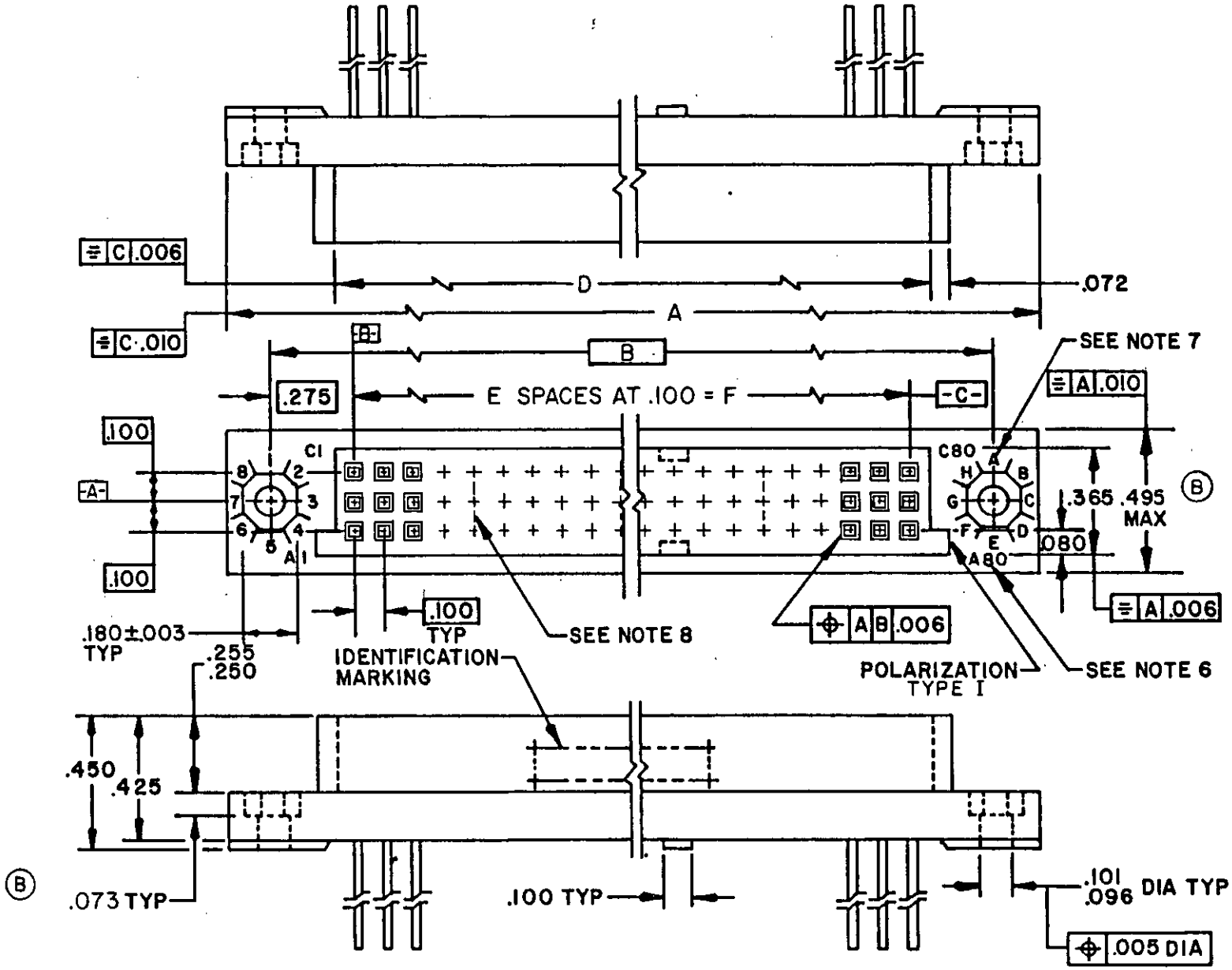
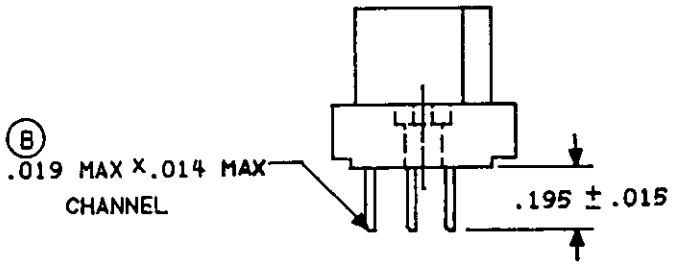
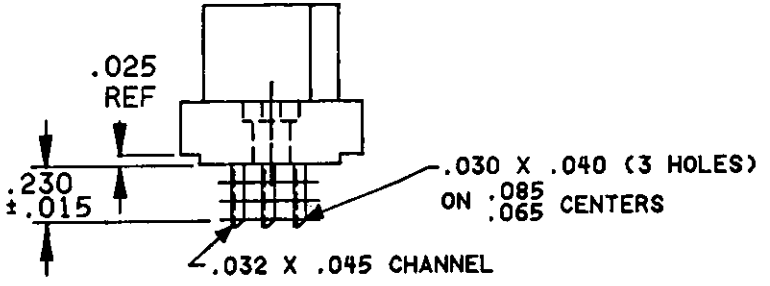


FIGURE 2. Connectors, receptacles (.100 spacing) 90 through 240 contact positions.

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FLOW SOLDER TYPE
CONTACT TYPE 1

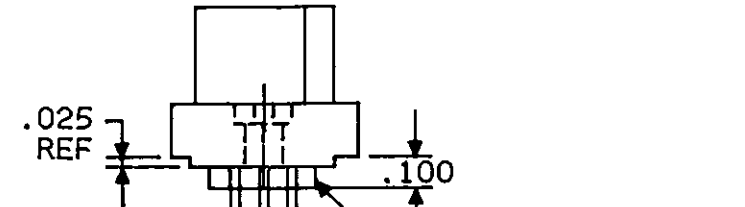


EYELET CONTACT
(SOLDER)
NO.26 AWG MAX

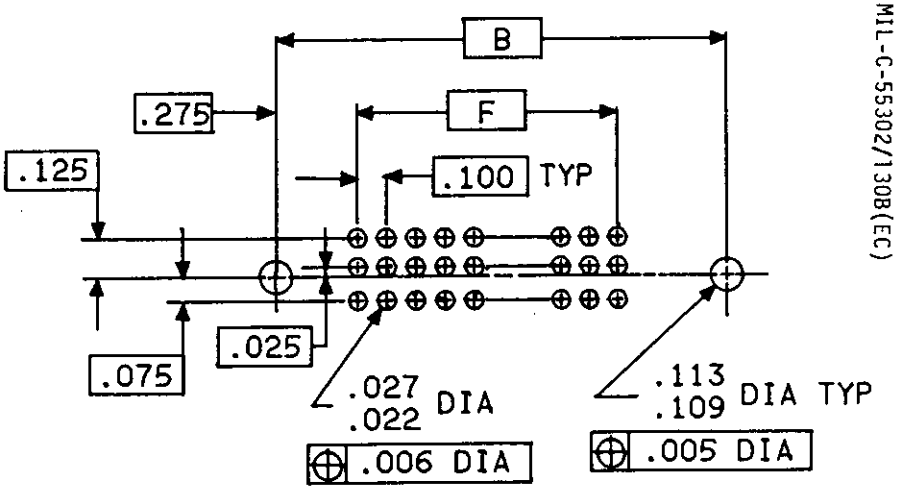
CONTACT TYPE 3

Inches	mm	Inches	mm	Inches	mm
.005	0.13	.027	0.69	.085	2.16
.006	0.15	.030	0.76	.100	2.54
.014	0.36	.032	0.81	.125	3.18
.015	0.38	.034	0.86	.195	4.95
.019	0.48	.040	1.02	.675	17.14
.022	0.56	.045	1.14		
.025	0.64	.075	1.90		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.



RACK MOUNT
WIRE WRAPPED TYPE
CONTACT TYPE 2



RECOMMENDED PRINTED CIRCUIT BOARD LAYOUT
CONNECTOR SIDE
FLOW SOLDER TYPE
CONTACT TYPE 1

FIGURE 2. Connectors, receptacles (.100 spacing)
90 through 240 contact positions - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figures 1 and 2, and tables I and II.

(B) Material: In accordance with MIL-C-55302.

(B) Plating: Types 1, 2, and 3 shall be nickel plated per MIL-C-55302. Over the nickel plate, the mating engagement area of types 1 and 3 for a minimum length of .100 inch and the entire contact type 2 shall be gold plated per MIL-C-55302. Over the nickel plate, the solder tail area for types 1 and 3 shall be tin-lead plated for .215 and .245 inch, respectively per MIL-C-55302.

Contact identification: See figures 1 and 2, and tables I and II.

(B) Current rating: 3.0 amperes, maximum per contact; 2.25 amperes continuous per contact at 75°F.

Keying (see MIL-C-55302/31):

Dash numbers -01 through -16: Two keys, part number M55302/31-08, and two key mounting hardware, part number M55302/31-09, packaged with connectors.

Dash numbers -17 through -46: Key, part number M55302/31-04, with .086-56 UNC-2A mounting screw is recommended.

Replacement contacts: See MIL-C-55302/32.

(B) Contact removal: See MIL-I-81969/9 for contact types 1 and 3. Use Amp 1-265871-7 tool for contact type 2.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by 0.25; the withdrawal force in pounds shall be a minimum of 0.025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding 0.020 ohm.

Contact retention: 3 pounds, minimum.

Dielectric withstanding voltage:

Sea level: 900 volts rms.

High altitude: 200 volts rms.

Resistance to test probe damage: The connector shall meet the requirements of MIL-C-55302/30.

Contact separation force: The connector shall meet the requirements of MIL-C-55302/29.

Oversized pin exclusion: .0465 diameter pin.

Part number: M55302/130- (and dash number from table I for 160 and 240 contact positions).

M55302/130- (and dash number from table II for 90 through 240 contact positions).

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

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TABLE I. Dash numbers, polarization types, and contact types for 160 and 240 contact positions.

Dash number	Number contacts	Type polarization	Type contact
01	160	I	1
02	160	I	2
03	160	II	1
04	160	II	2
05	160	III	1
06	160	III	2
07	160	IV	1
08	160	IV	2
09	240	I	1
10	240	I	2
11	240	II	1
12	240	II	2
13	240	III	1
14	240	III	2
15	240	IV	1
16	240	IV	2

TABLE II. Dash numbers and dimensions.
(For 90 through 240 contact positions)

Dash no.	No. of contacts	Type	A	B	D	E	F	Contact ident. no.
17 18 19	90	1 2 3	3.750 (95.25)	3.450 (87.63)	3.024 (76.81)	29	2.900 (73.66)	1-30
20 21 22	105	1 2 3	4.250 (107.95)	3.950 (100.33)	3.524 (89.51)	34	3.400 (86.36)	1-35
23 24 25	120	1 2 3	4.750 (120.65)	4.450 (113.03)	4.024 (102.21)	39	3.900 (99.06)	1-40
26 27 28	135	1 2 3	5.250 (133.35)	4.950 (125.73)	4.524 (115.57)	44	4.400 (111.76)	1-45
29 30 31	150	1 2 3	5.750 (146.05)	5.450 (138.43)	5.024 (127.61)	49	4.900 (124.46)	1-50
32 33 34	165	1 2 3	6.250 (158.75)	5.950 (151.13)	5.524 (140.31)	54	5.400 (137.16)	1-55
35 36 37	180	1 2 3	6.750 (171.45)	6.450 (163.83)	6.024 (153.01)	59	5.900 (149.86)	1-60
38 39 40	195	1 2 3	7.250 (184.15)	6.950 (176.53)	6.524 (166.37)	64	6.400 (162.56)	1-65
41 42 43	210	1 2 3	7.750 (196.85)	7.450 (189.23)	7.024 (178.41)	69	6.900 (175.26)	1-70
44 45 46	240	1 2 3	8.750 (222.25)	8.450 (214.63)	8.024 (203.81)	79	7.900 (200.66)	1-80

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